



Add Custom Slots to a Lex Chatbot



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▼ Slots (2) - optional [Info](#)

Information that a bot needs to fulfill the intent. The bot prompts for slots required for intent fulfillment, in priority order below.

Add slot

Q Filter

⋮

▶ Prompt for slot: accountType

Message: For which account would you like your balan...

Slot type

accountType

×

⋮

▶ Prompt for slot: dateOfBirth

Message: For verification purposes, what is your date ...

⚡ Slot type

AMAZON.Date

×



Introducing Today's Project!

In today's project, I used Amazon Lex to build and enhance a chatbot that can understand user input, collect required details through intents and slots, handle errors with fallback and failure responses, and guide users through checking their bank account balance in a natural, conversational way.

What is Amazon Lex?

Amazon Lex is a fully managed AWS service for building conversational interfaces using voice and text, and it is useful because it automatically understands natural language, manages conversation flow, and lets developers create intelligent chatbots without needing to build complex machine learning models from scratch.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how much fine-tuning goes into designing a smooth conversation flow, especially when configuring slot prompts and failure responses to handle invalid input without breaking the user experience.

This project took me...

This project took me 2 hours.



Slots

Slots are pieces of information that a chatbot needs to collect from the user in order to fulfill an intent, such as an account type or date of birth, and they help Lex guide the conversation and complete the user's request accurately.

By adding custom slots in utterances, my chatbot's users can speak or type naturally while BankerBot automatically extracts important details like their account type or date of birth without asking repetitive follow-up questions, making the conversation smoother, faster, and more user-friendly.

In this project, I have created custom slot types to capture specific user details such as bank account types so BankerBot can clearly understand valid inputs and accurately guide users through checking their account balance.



Slot type values

Modify the list of values used to train the machine learning model to recognize values for a slot.

Checking	Tab or ; or enter return for new value	×
Savings	Tab or ; or enter return for new value	×
Credit	Tab or ; or enter return for new value	×
	credit card	×
	visa	×
	mastercard	×
	amex	×
	american express	×

Add value

Maximum 140 characters. Valid characters: A-Z, a-z, 0-9, @, #, \$

☐ Use slot values as custom vocabulary [Info](#)



Connecting slots with intents

This slot type has restricted slot values, which means the chatbot will only accept predefined account types (and their synonyms) as valid input and will prompt the user again if they enter anything outside those allowed values.

I associated my custom slot with CheckBalance, which means BankerBot can now capture and store the user's account type and date of birth when they ask to check their balance, allowing the bot to collect the required details in the correct order before fulfilling the request.

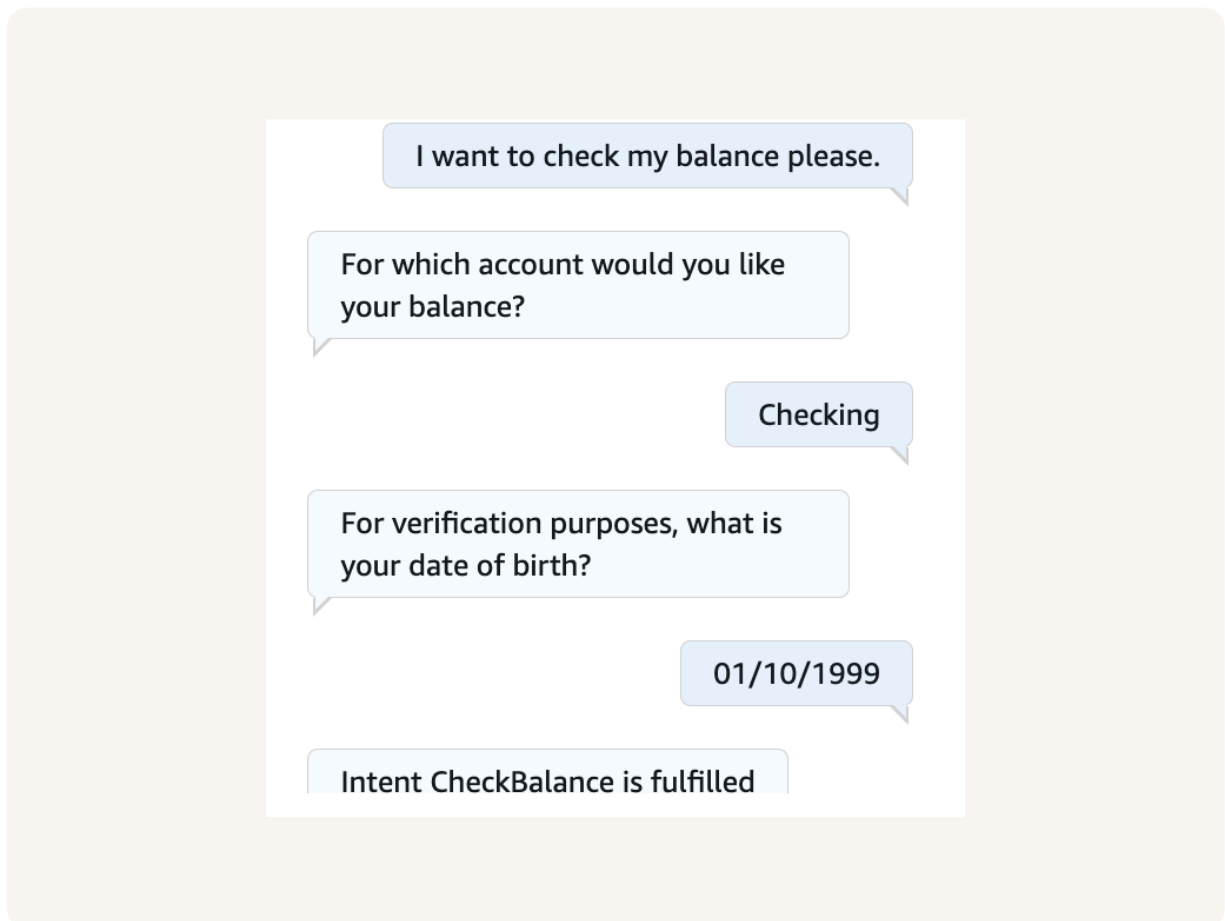
The screenshot shows the 'Slots' configuration page in the Amazon Lex console. At the top, it says '▼ Slots (2) - optional' with an 'Info' link. Below this is a description: 'Information that a bot needs to fulfill the intent. The bot prompts for slots required for intent fulfillment, in priority order below.' There is an 'Add slot' button in the top right. A search bar labeled 'Filter' is present. Two slots are listed:

Slot name	Slot type
Prompt for slot: accountType Message: For which account would you like your balan...	accountType
Prompt for slot: dateOfBirth Message: For verification purposes, what is your date ...	AMAZON.Date



Slot values in utterances

I included slot values in some of the utterances by adding placeholders for the slots directly into sample user phrases, so Amazon Lex knows where to extract the required information. For example, an utterance like “Check my {accountType} account balance” or “What’s my balance for my {accountType} account, my birthday is {dateOfBirth}” allows Lex to automatically capture those values from the user’s input.





Handling failures in slot values

I added variations for the `dateOfBirth` slot prompt, such as "Sorry, that wasn't clear to me. What's your date of birth?" and "Please try again as the previous attempt did not work". Because the messages play in order, my end user will first see the initial prompt, and only see the follow-up messages if their previous response was invalid, making the conversation clearer and more user-friendly.

I set up failure responses by configuring the `dateOfBirth` slot to handle invalid user input more intelligently. I also used failure responses to prompt the user to re-enter their birthday instead of ending the conversation or jumping to `FallbackIntent`, which keeps the flow focused on completing the `CheckBalance` request. A default setting I changed was the "Next step in conversation" option, switching it from moving to `FallbackIntent` to eliciting the same slot again (`dateOfBirth`) and enabling `Skip` elicitation prompt, so the chatbot waits for a valid date input without repeating the same question twice.



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what's YOUR birthday?!

For verification purposes, what is your date of birth?

i have no clue

Please try again as the previous attempt did not work.



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